PES University · Prompt Engineering Mini Project · 5th Semester AIML

ReviewIQ — Aspect-Based Customer Review Analytics with Structured Outputs

Final project report

Rachit S Mane (PES1UG24AM214) · P. Chinni Krishna Reddy (PES1UG24AM209) · Pranav V. Menon (PES1UG24AM198) · Akash R. (PES1UG24AM212)

Abstract

ReviewIQ converts unstructured Amazon appliance reviews into validated structured records (overall sentiment, aspect-level sentiment with verbatim evidence, and complaint categories) using a local large language model served by Ollama, with no cloud API. Outputs are validated with Pydantic and repaired automatically when invalid. On 299 evaluable reviews, three prompting strategies were compared with `qwen2.5:3b`; the highest overall-sentiment macro-F1 was 0.603 (Few-shot). A prompt chain aggregates extractions per product and generated cited summaries for all 15 products; a separate local LLM judge returned a valid verdict for only **5 of 15** summaries (**10 judge outputs failed** schema validation and have no score). In a negative control with deliberately corrupted summaries, the judge's faithfulness score decreased in **0 of 3** scored pairs, so its faithfulness scores are not a reliable quality measure. The reference labels are mostly AI-assisted (1 human, 298 AI-assisted), so the scores measure agreement with those labels; this and the other limitations are discussed in Section 10.

1. Introduction

Star ratings say how much customers liked a product but not why. Aspect-based sentiment analysis identifies what customers talk about (fit, installation, noise, durability, price) and how they feel about each aspect. Large language models can perform this analysis from free text, but their outputs are unreliable: they can produce invalid JSON, labels outside the taxonomy or evidence that does not appear in the review. ReviewIQ therefore treats the model as an unreliable component inside a validated pipeline.

2. Problem statement and objectives

"Develop an LLM pipeline that converts unstructured product reviews into validated structured records containing sentiment, product aspects and complaint categories, and generates per-product summaries. The pipeline shall use public e-commerce review data and compare prompting techniques for sentiment analysis and multi-label classification."

- **D1** — schema-based extraction with Pydantic and LangChain, JSON/CSV export and automatic repair of malformed outputs.
- **D2** — comparison of zero-shot, few-shot and Chain-of-Thought prompting on the 300-review labelled sample, with macro-F1.
- **D3** — prompt chain extract → aggregate → summarise with per-product summaries and LLM-as-a-Judge faithfulness evaluation.

3. Background

The dataset is the Amazon Reviews 2023 corpus (Hou et al., 2024). Prompting strategies follow the standard zero-shot, few-shot (in-context demonstrations) and Chain-of-Thought (explicit reasoning procedure) formulations. Evaluation uses macro-averaged F1, which weights classes equally and is therefore suited to the class-imbalanced sentiment labels. LLM-as-a-Judge uses a language model with a rubric to rate generated text; its known weaknesses (leniency, self-preference, sensitivity to fluency) motivate the separate judge model and deterministic checks used here.

4. System architecture

Reviews are processed by a strategy-specific prompt, sent to the local model through LangChain's `ChatOllama`, parsed, validated with Pydantic (including evidence grounding against the review text) and repaired when invalid. Valid records are aggregated per product with pandas; a summarisation prompt receives only the statistics and evidence quotes; a separate judge model scores each summary. All inference uses `http://127.0.0.1:11434`; non-local endpoints are rejected by configuration.

```
reviews → [EXTRACT] prompt → local LLM → parse → validate (+grounding) → repair loop → ReviewRecord / controlled failure
        → [AGGREGATE] pandas statistics + evidence quotes → [SUMMARISE] cited summary (validated ids) → [JUDGE] rubric 1–5 + deterministic checks
```

5. Methodology

5.1 Dataset

Amazon Reviews 2023, Appliances category (2,128,605 raw reviews), prepared deterministically (seed 42): text cleaning, 5–400 words, products with 200–5,000 valid reviews, 15 random products, de-duplication, 100 reviews per product — 1,500 reviews. The evaluation sample has 300 reviews stratified by star rating; the rating is never shown to the model or used as ground truth.

5.2 Annotation and label provenance

The reference labels contain 299 evaluable records: **1 human label** and **298 AI-assisted labels** generated by an AI model (Claude) from the review text only, following the annotation guide, with the project mentor's approval. One unreadable review is flagged and excluded. The provenance is recorded in the label file and shown in the application next to every metric.

5.3 Controlled comparison

All strategies used the same 299 review ids, model `qwen2.5:3b`, temperature 0.0, output cap 1024 tokens, context 8192 tokens, up to 2 repair attempts and strict evidence grounding. Few-shot demonstrations are synthetic, so they cannot overlap the test set. Chain-of-Thought performs its procedure silently and returns the same schema.

6. D1 — Structured extraction: implementation and results

Each review produces either a validated `ReviewRecord` or a controlled failure. The table summarises the extraction runs of the D2 experiment (the same records are exported as JSON and CSV; every exported record was reloaded and re-validated, including evidence grounding).

strategy	predicted	reviews	valid	failed (output)	failed (provider)	valid first try	needed repair	repaired OK	re-validated	means	hours	input tokens	output tokens
Chain-of-Thought	299	299	279	20	0	274	25	5	279/279	47.53	3.95	686970	66718
Few-shot	299	299	285	14	0	282	17	3	285/285	53.77	4.47	862456	68146
Zero-shot	299	299	282	17	0	273	26	9	282/282	41.19	3.42	615159	64126

7. D2 — Prompting strategy comparison

Status: **complete** · reference labels in run: 299 / 299 · missing predictions: {"zero_shot": 0, "few_shot": 0, "cot": 0}.

strategy	n	coverage	failed	accuracy	macro P	macro R	macro F1	weighted F1
Zero-shot	299	1.000	17	0.612	0.661	0.466	0.438	0.680
Few-shot	299	1.000	14	0.813	0.692	0.582	0.603	0.817
Chain-of-Thought	299	1.000	20	0.575	0.720	0.583	0.423	0.646

Complaints and aspects

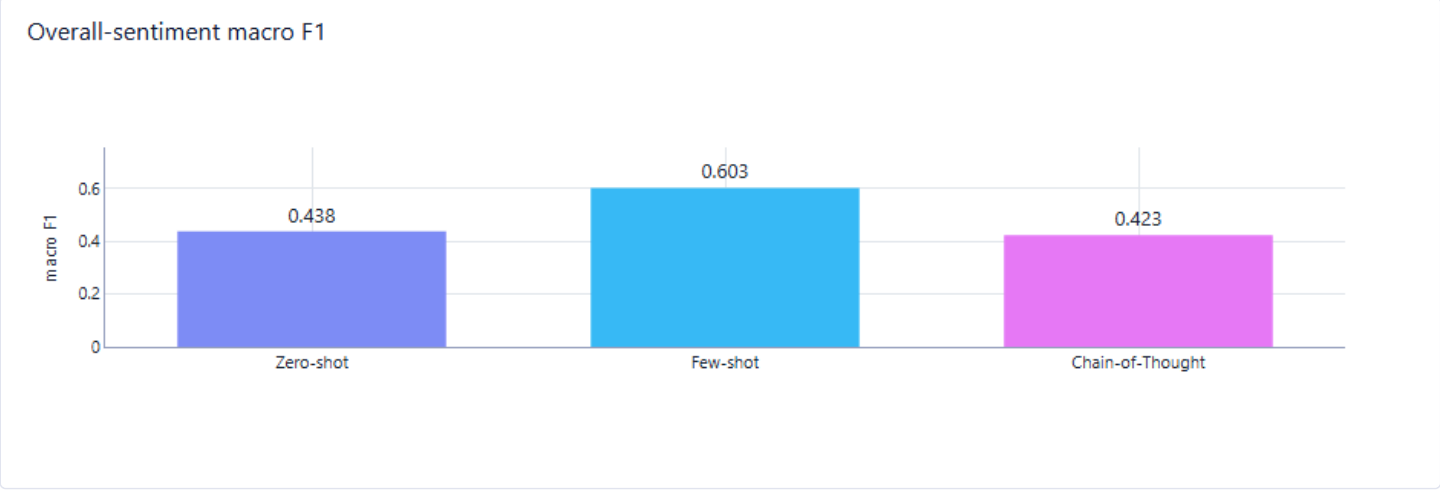
strategy	complaint micro F1	complaint macro F1	Hamming loss	aspect detection F1	aspect sentiment macro F1	joint F1
Zero-shot	0.359	0.303	0.071	0.522	0.551	0.394
Few-shot	0.415	0.360	0.077	0.649	0.648	0.558
Chain-of-Thought	0.385	0.285	0.069	0.503	0.552	0.370

Reliability and cost

strategy	validity	valid first try	needed repair	mean s / review	mean input tokens	mean output tokens
Zero-shot	0.943	0.913	26	41.2	2057	214
Few-shot	0.953	0.943	17	53.8	2884	228
Chain-of-Thought	0.933	0.916	25	47.5	2298	223

Per-class results (overall sentiment)

strategy	class	precision	recall	F1	support
Zero-shot	positive	0.952	0.531	0.682	111
Zero-shot	negative	0.868	0.837	0.852	141
Zero-shot	neutral	0.025	0.400	0.048	5
Zero-shot	mixed	0.800	0.095	0.170	42
Few-shot	positive	0.927	0.919	0.923	111
Few-shot	negative	0.830	0.901	0.864	141
Few-shot	neutral	0.143	0.200	0.167	5
Few-shot	mixed	0.867	0.309	0.456	42
Chain-of-Thought	positive	0.944	0.460	0.618	111
Chain-of-Thought	negative	0.883	0.801	0.840	141
Chain-of-Thought	neutral	0.053	1.000	0.101	5
Chain-of-Thought	mixed	1.000	0.071	0.133	42



Overall-sentiment macro-F1 per strategy.



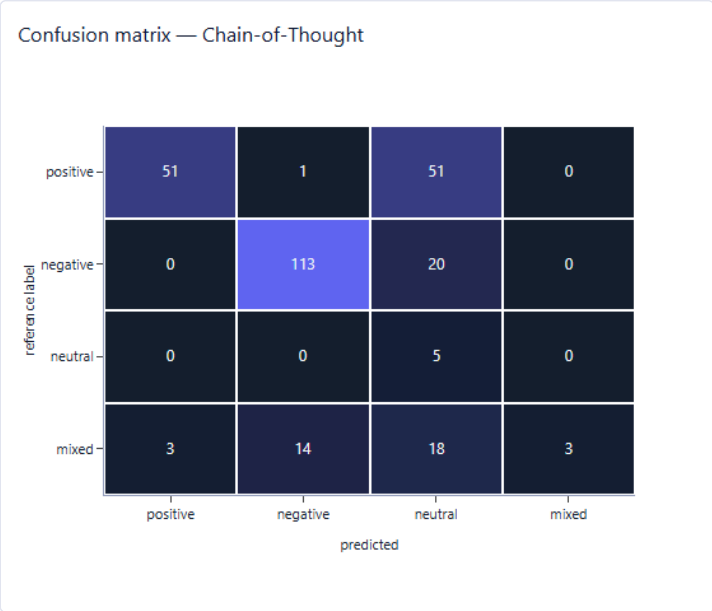
Per-class F1 per strategy.



Confusion matrix — Zero-shot (rows = reference, columns = prediction).



Confusion matrix — Few-shot (rows = reference, columns = prediction).



Confusion matrix — Chain-of-Thought (rows = reference, columns = prediction).

8. D3 — Prompt chain: extract → aggregate → summarise

> **Scope:** the chain was run on the **285 validly extracted reviews of the 299-review evaluation sample** (15 products, 9–42 reviews per product), **not on the full 1,500-review dataset**. Extractions were reused from the D2 run (source `dataset__few_shot__qwen2.5-3b__v1`) after a compatibility check of provider, model, prompt version and validation settings, so no review was sent to the model twice. Reviews whose extraction failed are not part of the aggregates.

15 products were summarised from their extracted reviews; 5 summaries were judged successfully and **10 judge call(s) failed** (the judge output was still invalid after the repair attempts; no score exists for those summaries and none was filled in). Summaries are model output and are kept with their statistics, evidence quotes and cited review ids.

product	judge	faithfulness	coverage	consistency	hallucination	mean	unsupported claims	citation polarity	top-aspect coverage	summary repairs
77L Deviled Egg Containers with Lid, (Set of 2),	failed (invalid judge output)	—	—	—	—	—	—	0.90	—	0
BLACK+DECKER BCED37 Compact Dryer for Standard W	failed (invalid judge output)	—	—	—	—	—	—	1.00	—	0
Dishwasher Silverware Basket	ok	4.0	4.0	3.0	1.0	3.0	0.0	1.00	1.00	0
Exact Fit 279834 Dryer Gas Valve Ignition Soleno	failed (invalid judge output)	—	—	—	—	—	—	1.00	—	0
GE WE1M964 Genuine OEM Timer Knob (Grey) for GE	failed (invalid judge output)	—	—	—	—	—	—	1.00	—	0
Napco Made in America 279838 AND 279816 279838 a	ok	4.0	4.0	3.0	2.0	3.25	0.0	1.00	1.00	0
Replace Samsung Heating Element DC47-00019A Drye	ok	3.0	3.0	2.0	2.0	2.5	1.0	0.71	1.00	0
SIMPLECUPS Disposable Cups for Use in Keurig Bre	failed (invalid judge output)	—	—	—	—	—	—	0.78	—	0
Solofill V1 GOLD CUP 24K Plated Refillable Filte	failed (invalid judge output)	—	—	—	—	—	—	1.00	—	0
Stanco 60555905 4 Pack Drip Bowl, GE/Hotpoint, C	failed (invalid judge output)	—	—	—	—	—	—	1.00	—	0
Tier1 ADQ36006101 Refrigerator Water & Air Filte	failed (invalid judge output)	—	—	—	—	—	—	1.00	—	0
UKF8001 Refrigerator Water Filter 4 Replacement	failed (invalid judge output)	—	—	—	—	—	—	1.00	—	0
Whirlpool 2198597 Icemaker, White	ok	4.0	4.0	4.0	1.0	3.25	0.0	1.00	1.00	0
Whirlpool 3949238 Washer Lid Switch	failed (invalid judge output)	—	—	—	—	—	—	1.00	—	0
Whirlpool 6-918873 Silverware Basket	ok	4.0	4.0	3.0	1.0	3.0	0.0	1.00	1.00	1

mean faithfulness 3.80 · mean coverage 3.80 · mean consistency 3.00 · mean hallucination 1.40 · summaries with all scores 5: 0 of 5.

Example summary — Whirlpool 3949238 Washer Lid Switch

Overall opinion (generated): The Whirlpool 3949238 Washer Lid Switch received a mixed to negative overall sentiment with 78.9% of reviews being negative and 15 out of 19 reviews containing complaints. The most frequently mentioned aspects were installation setup, performance, quality durability, price value, customer service, ease of use, and compatibility. The main complaints were related to poor durability (8 out of 15 reviews), installation difficulty (7 out of 15 reviews), poor value (6 out of 15 reviews), product defects (5 out of 15 reviews), customer service issues (4 out of 15 reviews), performance issues (4 out of 15 reviews), and compatibility fit issues (1 out of 15 reviews).

- 👍 Installation was easy and the part arrived on time. [R326a034e9581, R67a4f9906504]
- 👎 The product failed in less than 3 months. [R55ca2e9dc390]
- 👎 Installation was difficult and the part did not work as intended. [R326a034e9581, R037714d5ebd7]
- 👎 The product did not provide good value for money. [R28878e067e18]
- 👎 The product had issues with functionality and customer service. [R55ca2e9dc390, R037714d5ebd7]
- 👎 The product did not perform as expected and had compatibility issues. [R55ca2e9dc390, R28878e067e18]

Negative control: does the judge notice a corrupted summary?

Only the 5 summaries whose original judgement succeeded were eligible; 10 were skipped because their original judgement had failed (failed judgements are not re-sampled). Each eligible summary was copied and deliberately corrupted (overall opinion inverted with a fabricated percentage; invented feature: 'Reviewers praise the built-in Wi-Fi app control and voice-assistant support.'; all complaints removed) and judged with the same judge model (llama3.2) and prompt. Corrupted copies are stored separately and never shown as results.

	faithfulness	coverage	consistency	hallucination	mean
original (mean of 3)	4.00	4.00	3.33	1.33	3.17
corrupted (mean of 3)	4.00	3.33	3.00	1.33	2.92

Corrupted copy scored lower on the mean in **3 of 3** pairs, equal in 0, higher in 0; lower faithfulness in 0. The judge listed the invented feature as an unsupported claim in 0 and the fabricated percentage in 3 of 3 cases. 2 corrupted judgement(s) failed validation.

9. Evaluation, testing and verification

Independent recomputation of the D2 metrics

> Recomputed with the Python standard library only (no project code, no scikit-learn) from the raw prediction files and `data/annotations/annotations.json`, then compared with the stored `metrics.json` (tolerance 0.0001).
>
> **Result:** all recomputed values match the stored metrics (per-class P/R/F1 for all four classes, confusion matrices, macro P/R/F1, accuracy, invalid predictions, coverage, missing predictions; no duplicate or out-of-run ids).

> Metrics recomputed from the saved prediction files (no LLM calls) and compared with the stored `metrics.json`.
>
> ## `20261111-030912_qwen2.5-3b`
>
> - status: **complete** · reference labels in run: 299 / 299
> - missing predictions: {'zero_shot': 0, 'few_shot': 0, 'cot': 0}
> - recomputation matches stored metrics: **yes**
>
> - D1 export copied from `outputs\live\d1\20261111-030912_qwen2.5-3b`
> - D3 export copied from `outputs\live\d3\dataset__few_shot__qwen2.5-3b__v1`
> - Judge negative control copied from `outputs\live\d3\judge_sanity\dataset__few_shot__qwen2.5-3b__v1`
>
> - All 7 JSON and 7 CSV files under `reports/` parse successfully.

Discussion

D2 — what the comparison shows

- **Few-shot is clearly best** on the primary metric: overall-sentiment macro-F1 **0.6025**, against 0.4380 for zero-shot and 0.4232 for Chain-of-Thought (accuracy 0.8127 vs 0.6120 and 0.5753). It is also best on aspect detection (micro-F1 0.6488 vs 0.5221 / 0.5028), aspect sentiment (macro-F1 0.6479), joint aspect-sentiment pairs (0.5579) and complaint micro-F1 (0.4153), and it has the highest output validity (95.3%) and first-attempt validity (94.3%).
- **The main failure mode of zero-shot and CoT is over-predicting `neutral`.** Zero-shot predicted neutral for 79 reviews and CoT for 94, although only 5 reference labels are neutral; 44 (zero-shot) and 51 (CoT) reference-positive reviews were labelled neutral. Few-shot predicted neutral only 7 times. The demonstrations calibrate what "neutral" means far better than the written label definitions alone.
- **`mixed` is the hardest class for every strategy** (recall 0.095 / 0.310 / 0.071). Mixed reviews are most often predicted as negative (few-shot: 19 of 42), i.e. the model lets the complaint dominate.
- **The CoT procedure did not help this 3B model.** It raised macro precision (0.7201) only because it rarely predicts mixed, while recall on positive fell to 0.4595. Its silent reasoning instructions add prompt length without improving calibration; it also produced the most output failures (20).
- **Cost:** few-shot is the most expensive per review (mean 2,884.47 input tokens, 53.8 s) because of the five demonstrations; zero-shot is the cheapest (2,057 tokens, 41.2 s). On this hardware the accuracy gain is worth the ~30% extra time.
- **Failures are counted, not hidden:** 17, 14 and 20 reviews had no valid output after two repair attempts; they are scored as wrong. Final failure causes (zero-shot / few-shot / CoT): evidence not verbatim 6 / 5 / 8, out-of-taxonomy label 5 / 5 / 6, evidence longer than 300 characters 4 / 1 / 1, invalid or truncated JSON (including runaway generation at the 1,024-token cap) 0 / 3 / 2, missing or extra fields 2 / 0 / 0, over-long rationale 0 / 0 / 3.

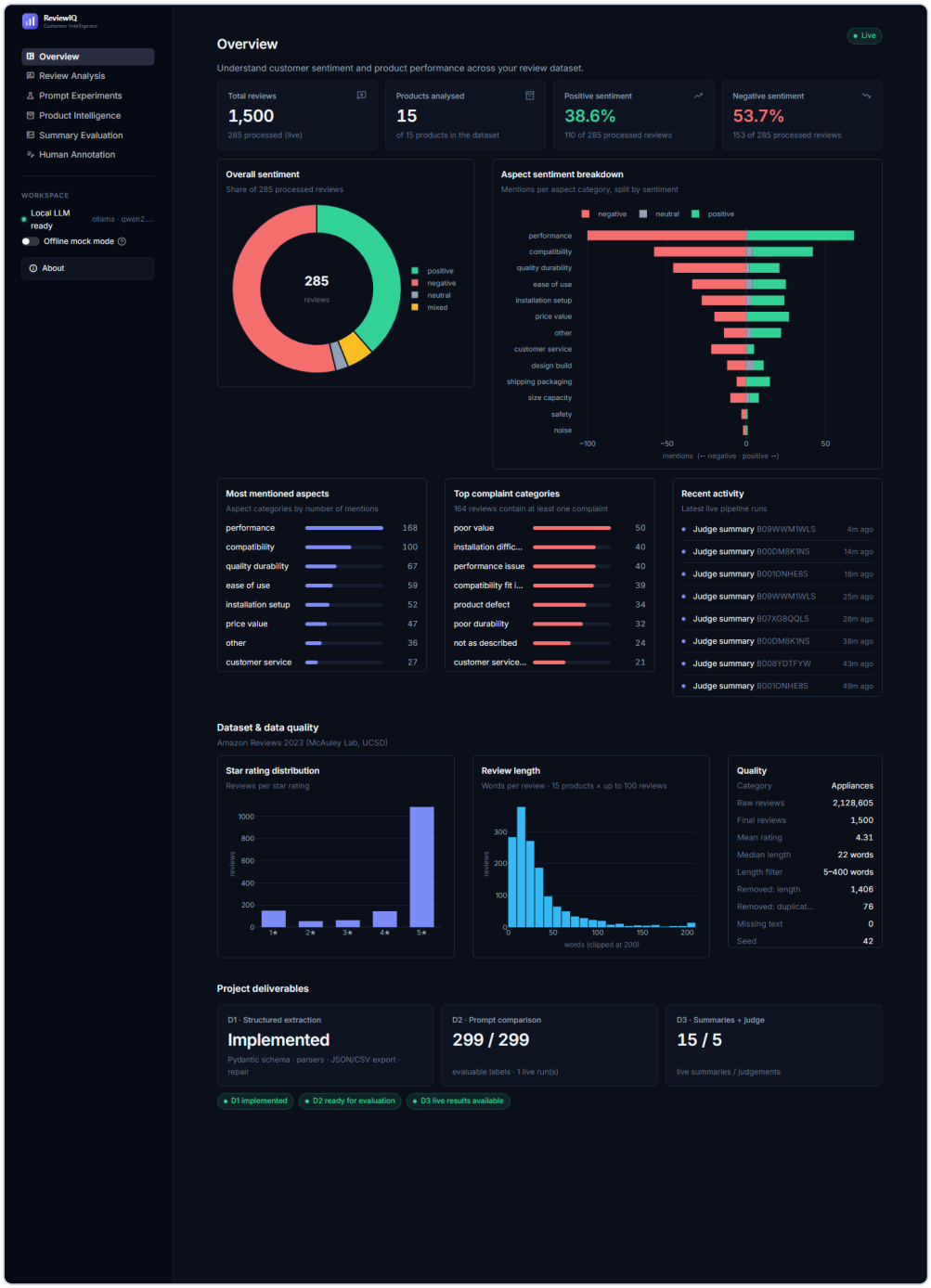
Reading the numbers correctly

- 298 of the 299 reference labels are AI-assisted; the scores measure agreement with those labels.
- `neutral` has only 5 reference examples, so its per-class F1 is very noisy and pulls macro-F1 down for all strategies.
- The sample is rating-stratified (about 60 reviews per star rating), so accuracy describes this balanced set, not the natural 72%-five-star distribution of the dataset.

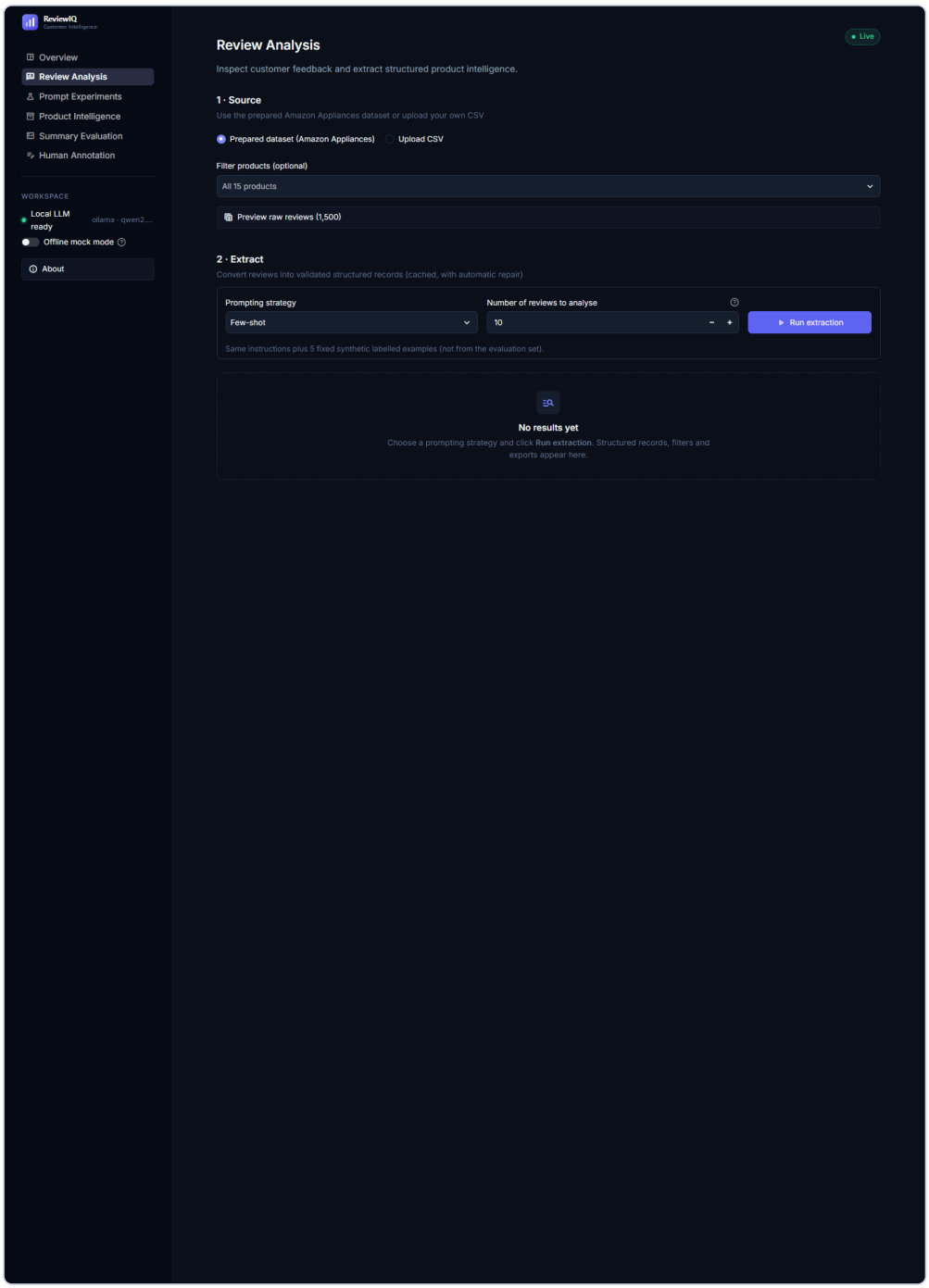
D3 — what the chain and the judge show

- **The summariser was reliable; the judge was not.** qwen2.5:3b produced a valid, citation-grounded summary for all 15 products (14 on the first attempt, 1 after one repair). The local judge (llama3.2) returned a schema-valid verdict for only **5 of 15** summaries; the other 10 were still invalid after two repair attempts — it either gave a bare score per dimension instead of a score plus justification, or nested the `unsupported_claims` list inside every dimension. Those 10 summaries have no judge score; none was filled in, and the failures were not re-sampled.
- **The judge's scores are internally inconsistent.** In the 5 valid verdicts faithfulness is 3–4 while hallucination is 1–2 (1 = "many unsupported claims"), so the two dimensions contradict each other.
- **Negative control: the judge did not detect unfaithful content.** Of the 5 eligible pairs, 3 produced valid verdicts for the corrupted copy (2 corrupted judgements failed validation). The corrupted copy's mean score was lower in all 3 pairs, but only through coverage and consistency (it noticed the removed complaints); **faithfulness was lower in 0 of 3** and the invented "Wi-Fi app / voice assistant" feature was flagged in 0 of 3. The fabricated percentage was flagged in 3 of 3, which the deterministic percentage check also catches.
- **How to read this:** with 3 scored pairs the control is far too small for any statistical claim; it is evidence *against* using this 3B judge's faithfulness and hallucination scores as a quality measure. The deterministic checks (cited ids belong to the product, percentages appear in the statistics, citation polarity, top-aspect coverage) are the more trustworthy D3 evidence in this project. A larger judge or human rating of the summaries would be needed to evaluate faithfulness properly.

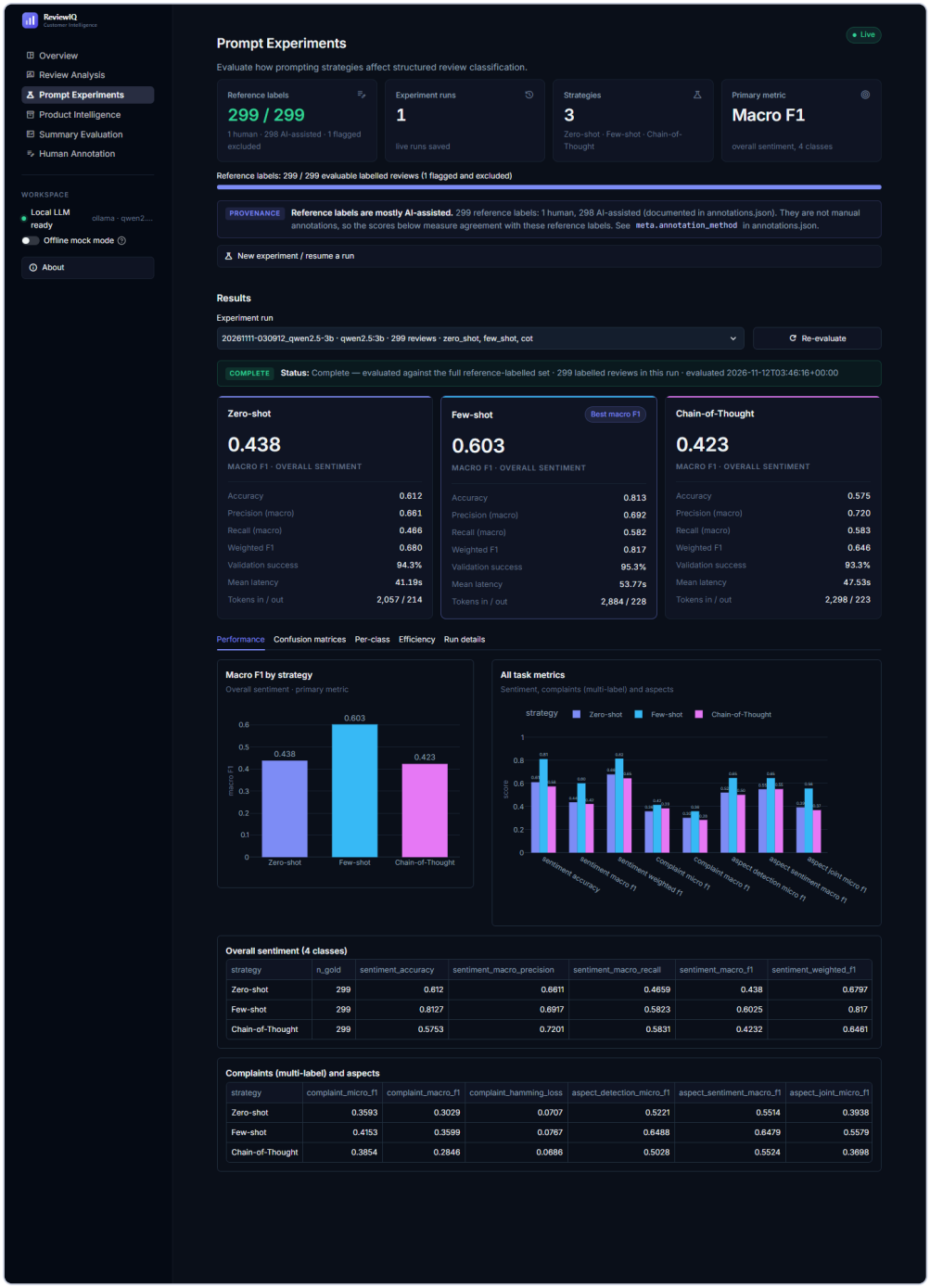
Screenshots (captured from the running application)



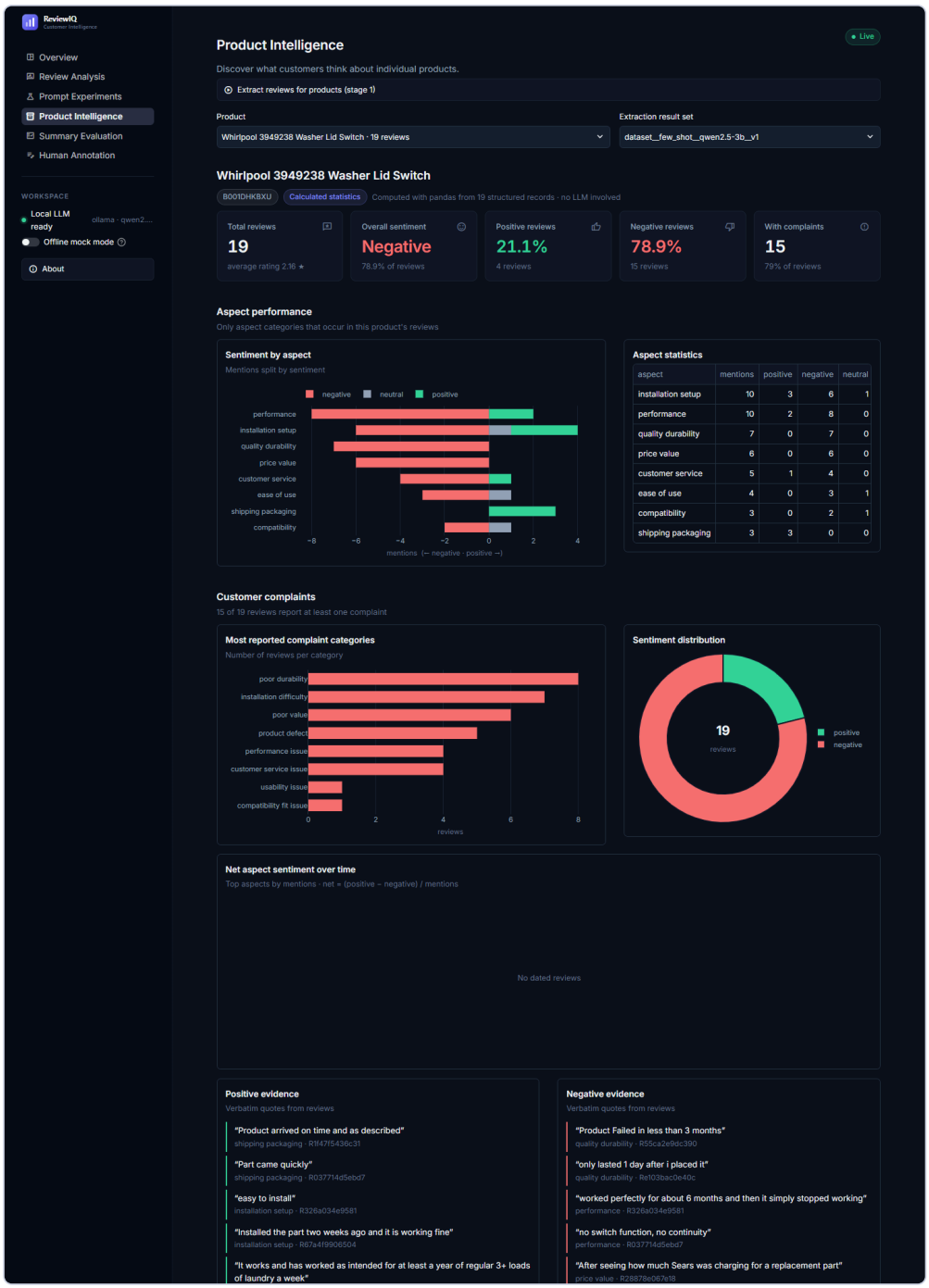
Overview — dataset size, deliverable status and processed reviews



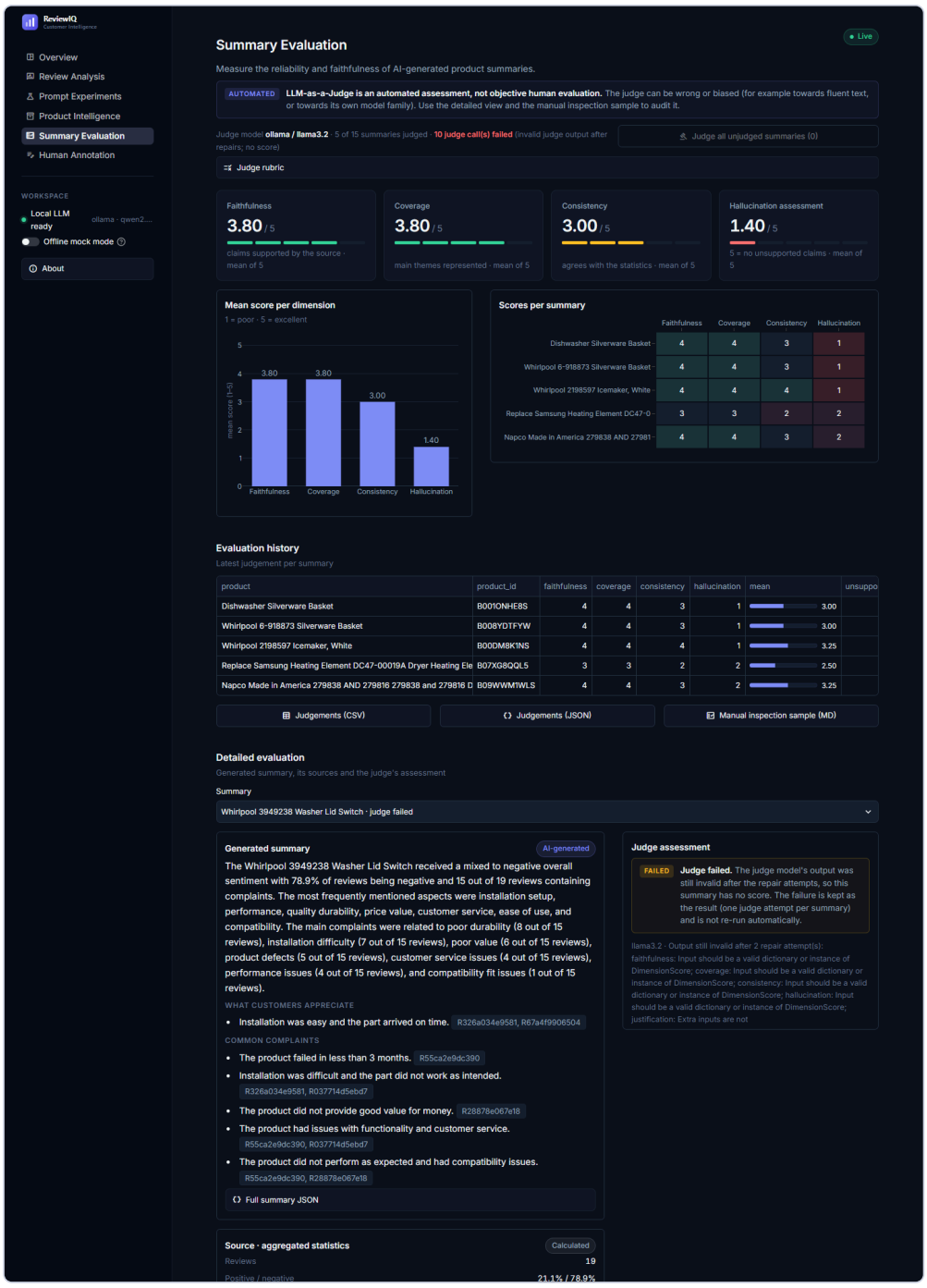
Review Analysis — the extraction page in its initial state (data source, prompting strategy and run controls); no extraction was run in this browser session, so no records are shown



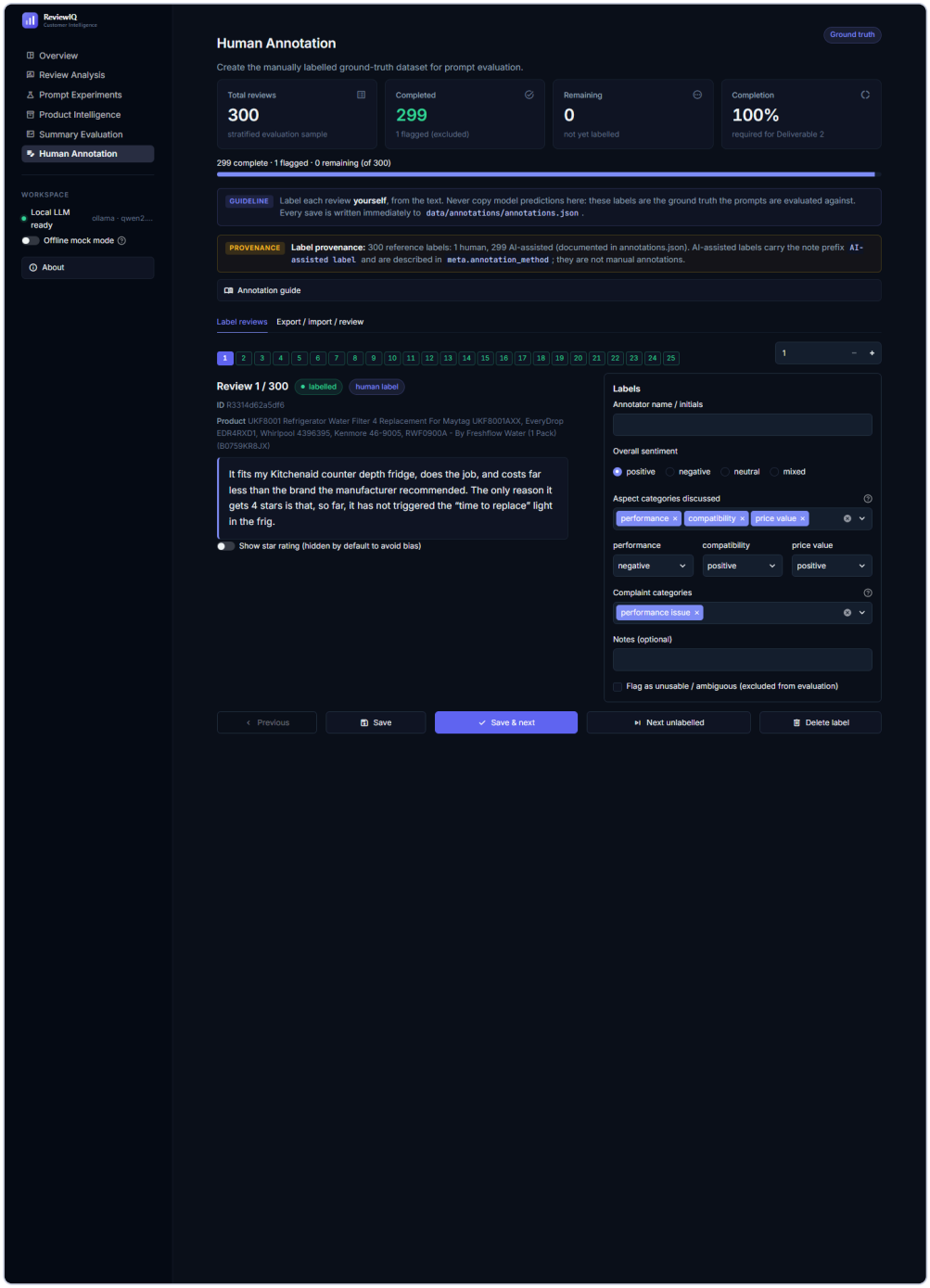
Prompt Experiments — measured comparison of the three strategies



Product Intelligence — calculated statistics, aspect performance, complaint categories and evidence quotes for one product; the generated summary lower on the page is outside the captured area



Summary Evaluation — LLM-as-a-Judge scores and deterministic checks



Human Annotation — reference labels with provenance

10. Limitations

- 298 of the 299 evaluable reference labels are AI-assisted; the scores measure agreement with these labels, not independent manual annotation, and there is no inter-annotator agreement.
- The models are small enough for a 4 GB GPU. The 3B model misses aspects and complaints. The 7B model only partly fits in VRAM (about 88 s per review in the smoke test), so no 7B evaluation on the 299 reviews was run; only a one-review smoke test exists.
- The 3B model occasionally loops; a 1,024-token cap truncates such outputs, which are then repaired or counted as failures.
- Aspect matching is at category level; the evaluation sample is rating-stratified, so accuracy describes the balanced set.
- LLM-as-a-Judge scores come from a 3B model and are not calibrated against human ratings. D3 product summaries are based on the evaluation sample (9–42 extracted reviews per product), not on all 100 reviews per product.

11. Conclusion

ReviewIQ demonstrates a complete, locally hosted pipeline that turns free-text reviews into validated structured data, compares prompting strategies under controlled conditions with transparent coverage and failure accounting, and produces cited product summaries for all 15 products. The LLM-as-a-Judge stage is auditable but did not work reliably with the small local judge: only 5 of 15 judge outputs were valid and 10 failed, and in the negative control the judge's faithfulness score decreased in 0 of 3 scored pairs (the invented feature was flagged in 0 of 3), with too few pairs for a statistical conclusion. The deterministic checks are the more trustworthy D3 evidence. The validation and repair layer makes even a small local model usable for structured extraction, while the measured results show the limits of 3B-parameter models for fine-grained aspect and complaint extraction.

References

1. Hou, Y., Li, J., He, Z., Yan, A., Chen, X., & McAuley, J. (2024). *Bridging Language and Items for Retrieval and Recommendation*. arXiv:2403.03952. (Amazon Reviews 2023 dataset)
2. Project repository: https://github.com/RachitSMane/Prompt-engineering-mini-project-
3. Software: Ollama, LangChain (langchain-core, langchain-ollama), Pydantic, pandas, scikit-learn, Streamlit, Plotly — versions in reports/execution_log.md.